

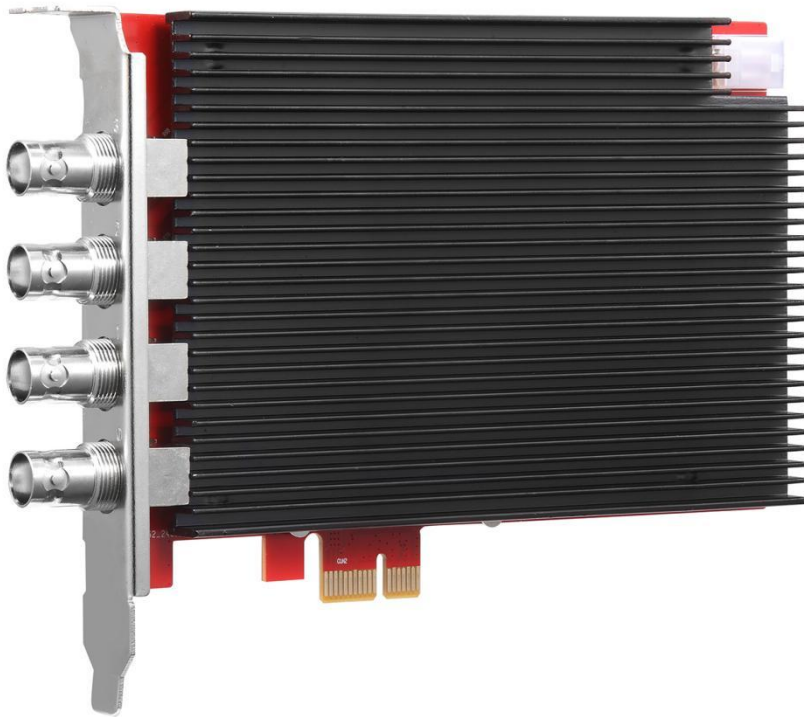


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TBS6324 Quad 3-G SDI PCIe Encoder User Manual for Linux



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SAFETY NOTES

Read the user manual before using the equipment, mainly "**SAFETY RULES**" paragraph.
The symbol on the equipment means "**SEE USER MANUAL**". In this manual may also appear as a Caution or Warning symbol.

WARNING AND CAUTION statements may appear in this manual to avoid injury hazard or damage to this product or other property.

USER MANUAL VERSION

Version	Date
1.0	Oct 2024

SAFETY REQUIREMENTS

- * **The security can be compromised if not applied the instructions in this manual.**
- * Use this instrument under the **specified environmental conditions**.
- * The user is not allowed to perform changes inside the equipment.
Any change on the equipment must be done exclusively by specialized staff.
- * Use appropriate low-level radiation cables for input / output signals, especially on high level signals.
- * Follow the **cleaning instructions** described in the Maintenance paragraph.

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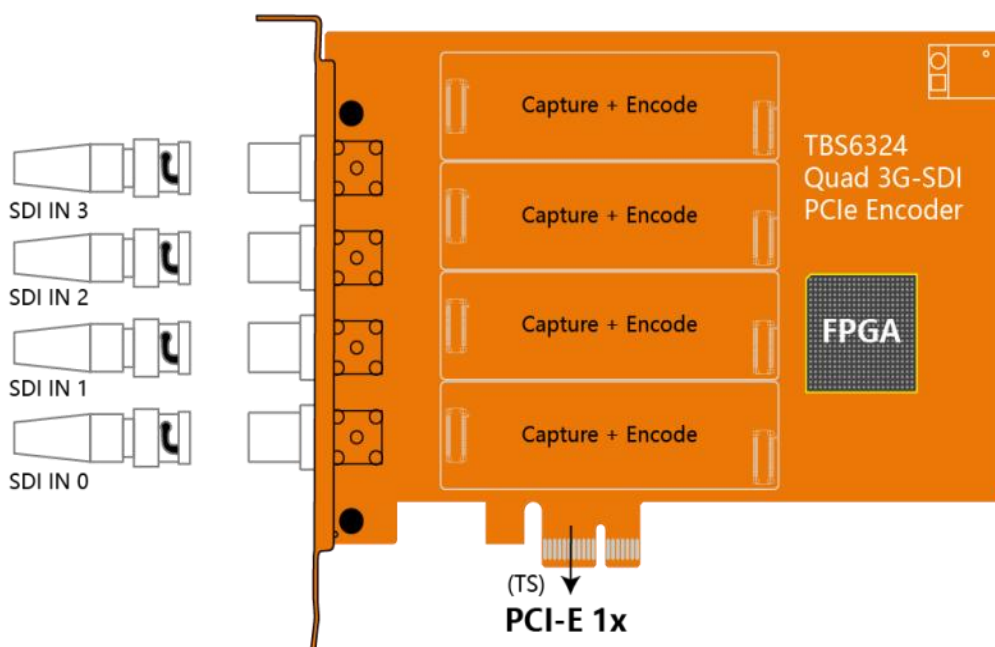
1. Product Introduction

1.1 Overview

TBS6324 quad 3G-SDI PCIe encoder allows capturing and encoding four separate 3G/HD/SD-SDI sources simultaneously. It supports up to 1080P@60HZ input and H.265/H.264 1080P@60fps video encode. It is an ideal device for those who need to stream lower bit rate video over LAN or WAN networks, often due to internal network constraints, while still maintaining high quality video.

After capturing 3G/HD/SD video from SDI ports, TBS6324 will generate H.264/H.265-encoded stream in TS packet and output via PCIe interface. TBS6324 is based on DVB architecture, so it is compatible with various DVB software such as Kylone, TVheadend, Astra, Cesbo, DVB, Dream, DVBCViewer, TSReader etc.

TBS6324 provides versatile and cost-efficient video encoding solution in various industries. It can be used for live production, live streaming, broadcast TV, remote transmission, post-production etc.



1.2 Specification

Model NO.	TBS6324
Model Name	Quad 3G-SDI PCIe Encoder
Input	4x SDI (BNC Female)
Interface	PCIe 1x
Output	TS (Transport stream)
Video	
Video Input	SD-SDI (SMPTE 259): 480i, 576i
	HD-SDI (SMPTE 292): 1920x1080i_60/50hz, 1920x1080p_30/25/24hz, 720p_60/50hz
	3G-SDI (SMPTE 242): 1920x1080p_60/50hz Level A & Level B
Video Encode Size	auto, 1920x1080, 1680x1056, 1280x720, 1024x576, 720x576, 720x480, 704x576, 640x480, 640x360
Video Output	1920x1080p@60/50/30/25/24fps, 1280x720p@60/50/30/25fps, 720x576p@30fps, 640x480p@30fps etc.
Video Codec	H.264, H.265
Profile	H.264: Baseline, Main, High H.265: Main
Level	H264: Level 4.2; H265: Level 4.1
Bitrate Control	CBR, VBR, EVBR
Frame Rate	5~60fps
Key Interval (GOP)	5~180fps
GOP Structure	IPPP
Bitrate	300~20000Kbps
Color depth	8 bit
Chroma subsampling	4:2:0
De-interlace	Supported, force to do de-interlace when i (interlace) input
Audio	
Audio Input	SDI embedded
	Supports up to 4 stereo audio tracks (8 mono audio channels): channel 1-2, 3-4, 5-6, 7-8
Audio Codec	AAC-LC, AAC-HE
Audio Bitrate	24, 32, 44.1, 48, 64, 96, 128, 160, 192, 256(Kbps)
Audio Resample	48, 44.1, 32(Khz)
System & Software	
System	Linux/Windows 64-bit/32-bit driver
Requirements	Available PCI Express x1, x4, x8 or x16 slot

Control	Linux command line setting tool
	Windows GUI setting tool
Software	Compatible with various DVB software
	Linux platform: DVBlaster, Astra Cesbo, Tvheadend etc.
	Windows: DVBdream, DVBviewer, TSReader, TransEdit etc.
General Spec.	
Power Consumption	12 Watt
Net Weight	290g
Size	141mm x 104.5mm
Working Environment	Operating temperature: 0 to 60 deg °C
	Storage temperature: -20 to 70 deg °C
	Relative Humidity: 5% to 90% (non-condensing)
Package Content	1x TBS6324
	1x DC line

2. Hardware Installation and Requirements

➤ Hardware installation:

Install the TBS6324 Quad 3-G SDI PCIe Encoder card into a Desktop PC or Server that has a PCIe slot, ensuring that the card is tightly secured in the PCIe slot. The TBS6324 features a PCIe 1x & GEN 1 interface and is also compatible with PCIe 4x/8x/16x slots.

➤ Desktop PC/Server Hardware configuration requirements:

Intel 4th generation i3 CPU, or a more powerful Desktop PC/Server.
4GB RAM or higher

➤ Additional Notes:

This is a Hardware Encode card, which uses an encode chipset to handle the “encode” processing. Therefore, it does not consume PC/Server CPU resources.

3. Detection and Power Supply

➤ Check whether card is detected/recognized by Desktop PC or Server:

To verify if the card is detected/recognized by Desktop PC or Server, use the following command:

```
# lspci -vvv | grep 6324
```

```

root@dw3-All-Series:/home/dw-3/TBS6324SDItool#
root@dw3-All-Series:/home/dw-3/TBS6324SDItool#
root@dw3-All-Series:/home/dw-3/TBS6324SDItool# lspci -vvv | grep 6324
pcilib: sysfs_read_vpd: read failed: No such device
pcilib: sysfs_read_vpd: read failed: No such device
Subsystem: Device 6324:0010
root@dw3-All-Series:/home/dw-3/TBS6324SDItool#
root@dw3-All-Series:/home/dw-3/TBS6324SDItool#
root@dw3-All-Series:/home/dw-3/TBS6324SDItool#
root@dw3-All-Series:/home/dw-3/TBS6324SDItool#
root@dw3-All-Series:/home/dw-3/TBS6324SDItool#
root@dw3-All-Series:/home/dw-3/TBS6324SDItool#

```

- If the `lspci` command lists the device, it means the card has been detected/recognized and the hardware connection is ready. However, if the device is not listed, it indicates that the PC/Server did not detect the card. In this case, please check the hardware installation again. The card may not be installed tightly or there may be an issue with the PCIe slot. As a troubleshooting step, try installing the card in another PCIe slot or using a different Desktop PC/Server to test.
- Typically, the PCIe Encoder is powered by the PCIe slot itself, eliminating the need for an additional power connection. However, there are instances where special server PCIe slots may be unable to provide power, or they may not be capable of supplying sufficient power for the encoder. In such cases, connecting an additional power source will be necessary.
- **Power Consumption:**
The power consumption of this PCIe Encoder is approximately 12 watts.

4. Linux Driver Installation

- **TBS Linux official open source driver:**

```

1  # Installing Linux Drivers
2
3  ## TBS Linux Official Open Source Drivers
4
5  Driver repository:
6  [TBS Linux Media GitHub Page](https://github.com/tbsdtv/linux_media/wiki)
7
8  To download the drivers:
9  ```bash
10 # git clone https://github.com/tbsdtv/media_build.git
11 # git clone --depth=1 https://github.com/tbsdtv/linux_media.git -b latest ./media

```

https://github.com/tbsdtv/linux_media/wiki

- **download driver:**
`git clone https://github.com/tbsdtv/media_build.git`

```
# git clone --depth=1 https://github.com/tbsdtv/linux_media.git -b latest ./media
```

➤ **Install some basic tools, dependencies:**

For CentOS system:

bash

```
1 # sudo yum install centos-release-scl
2 # sudo yum install devtoolset-8-gcc*
3 # scl enable devtoolset-8 bash
4 # mv /usr/bin/gcc /usr/bin/gcc-bkk
5 # mv /usr/bin/g++ /usr/bin/g++-bkk
6 # ln -s /opt/rh/devtoolset-8/root/usr/bin/gcc /usr/bin/gcc
7 # ln -s /opt/rh/devtoolset-8/root/usr/bin/g++ /usr/bin/g++
```

For Ubuntu system:

bash

```
1 # sudo apt-get update
2 # sudo apt-get install gcc-8
3 # sudo apt-get install g++-8
4 # cd /usr/bin
5 # sudo rm gcc g++
6 # sudo ln -s gcc-8 gcc
7 # sudo ln -s g++-8 g++
```

```
# sudo apt-get update
# sudo apt-get install gcc-8
# sudo apt-get install g++-8
# cd /usr/bin
# sudo rm gcc g++
# sudo ln -s gcc-8 gcc
# sudo ln -s g++-8 g++
```


Starts to install driver (build the driver):

bash

```
1 # cd media_build
2 # make dir DIR=../media
3 # make allyesconfig
4 # sed -i -r 's/(^CONFIG.*_RC.*=)../\1n/g' v4l/.config
5 # sed -i -r 's/(^CONFIG.*_IR.*=)../\1n/g' v4l/.config
6 # make -j4
7 # sudo make install
```

```
# cd media_build
# make dir DIR=../media
# make allyesconfig
# sed -i -r 's/(^CONFIG.*_RC.*=)../\1n/g' v4l/.config
# sed -i -r 's/(^CONFIG.*_IR.*=)../\1n/g' v4l/.config
# make -j4
# sudo make install
```

➤ **Note:**

If any error occurs during the "build/compile" process, please fix the error first and then continue with the installation.

➤ **Common Build Errors and Solutions:**

For a list of frequently encountered errors and their corresponding solutions, please refer to: https://github.com/tbsdtv/linux_media/wiki/Prerequisites,-Build-errors-and-Rebuilding
After finishing the driver installation, please reboot the desktop PC/server to load the encoder devices.

➤ If the driver is installed correctly, the encoder devices will load as follows:

```
root@dw3-All-Series:/home/dw-3/TBS6324SDItool#
root@dw3-All-Series:/home/dw-3/TBS6324SDItool# ls /dev/dvb
adapter0 adapter1 adapter2 adapter3
root@dw3-All-Series:/home/dw-3/TBS6324SDItool#
root@dw3-All-Series:/home/dw-3/TBS6324SDItool#
root@dw3-All-Series:/home/dw-3/TBS6324SDItool#
root@dw3-All-Series:/home/dw-3/TBS6324SDItool#
root@dw3-All-Series:/home/dw-3/TBS6324SDItool#
root@dw3-All-Series:/home/dw-3/TBS6324SDItool# dmesg | grep frontend
[ 7.483783] begin Attaching frontend
[ 7.592294] end Attaching frontend
[ 7.592313] TBSECP3 driver 0000:06:00.0: DVB: registering adapter 0 frontend 0 (TurboSight TBS 6324 3G-SDI PCIe Encoder)...
[ 7.714944] begin Attaching frontend
[ 7.820310] end Attaching frontend
[ 7.820329] TBSECP3 driver 0000:06:00.0: DVB: registering adapter 1 frontend 0 (TurboSight TBS 6324 3G-SDI PCIe Encoder)...
[ 7.938896] begin Attaching frontend
[ 8.044310] end Attaching frontend
[ 8.044329] TBSECP3 driver 0000:06:00.0: DVB: registering adapter 2 frontend 0 (TurboSight TBS 6324 3G-SDI PCIe Encoder)...
[ 8.166967] begin Attaching frontend
[ 8.272281] end Attaching frontend
[ 8.272303] TBSECP3 driver 0000:06:00.0: DVB: registering adapter 3 frontend 0 (TurboSight TBS 6324 3G-SDI PCIe Encoder)...
root@dw3-All-Series:/home/dw-3/TBS6324SDItool#
```

Note: There are a total of four adapters.

5. Encoding Settings

- For the TBS6324 Quad 3G-SDI PCIe Encoder, we provide a highly efficient command-line setting tool that allows for quick configuration. This tool supports setting one encoding parameter at a time and also allows for the simultaneous configuration of multiple encoding parameters.

```
root@dw3-All-Series:/home/dw-3/TBS6324SDITool#
root@dw3-All-Series:/home/dw-3/TBS6324SDITool#
root@dw3-All-Series:/home/dw-3/TBS6324SDITool#
root@dw3-All-Series:/home/dw-3/TBS6324SDITool# ls
bin  logo1.bmp  readme  src  TBS6324SDITool_x64  upgrade.bin  za
root@dw3-All-Series:/home/dw-3/TBS6324SDITool#
root@dw3-All-Series:/home/dw-3/TBS6324SDITool#
root@dw3-All-Series:/home/dw-3/TBS6324SDITool#
root@dw3-All-Series:/home/dw-3/TBS6324SDITool#
```

- The setting tool 'TBS6324SDITool_x64' has been pre-compiled for Ubuntu systems, allowing to run the binary directly. If using a different Linux distribution, need to compile the tool from source.

- **Compiling the Setting Tool:**

The source code for the setting tool is located in the /src/ directory. To compile the tool, follow these steps:

```
# cd ./src
# make clean
# make
```

After successful compilation, the setting tool will be located in the /bin/ directory.

- **Using the Setting Tool:**

The command line tool allows to configure all encode parameters, including:

Output video size

- Video Codec (H.265/H.264)
- Frame rate
- Key interval (key frame)
- Bitrate control
- Video bitrate
- TS PIDs
- Audio Codec (AAC_LC)
- Audio bitrate
- Audio re-sample

- And other parameters

```

root@dw3-All-Series:/home/dw-3/TBS6324SDITool# ./TBS6324SDITool_x64

The version of the tool is V1.0.0.1
-a : <adapter>
-s : <encoded size: 0(auto),1(640x360),2(640x480),3(704x576),4(720x480),5(720x576),6(1024x576),7(1280x720),8(1680x1056),9(1920x1080)>
-f : <enc_fps: 5-60>
-e : <enc_type: H264 or H265>
-l : <codec_type: cbr,vbr,evbr>
-b : <cbr,vbr and evbr bitrate(Kbps): 160-20000>
-t : <pmt,transpid,vpid,sid>: pmt(16-7936), transpid and vpid(16-3840)
-N : <program_name> set the program Name at most 16 characters
-k : <venc_gop: 1-180 (key interval/key frame under IPPP)>
-q : <minQp: 5-48 (activated when sets to vbr or evbr bitrate control)>
-Q : <maxQp: minQp-51 (activated when sets to vbr or evbr bitrate control)>
-p : <H264 profile: 0:baseline, 1:main, 2:high; H265:supports main profile only, force encode to main profile>
-T : <ts null packet: 0(off),2(1.2x),3,4,5,6,7,8,9,10(2.0x),12,14,16,18,20(3.0x)>
-I : <audio bitrate(bps): 24000,32000,44100,48000,64000,96000,128000,160000,192000,256000>
-A : <AAC type: LC-AAC, HE-AAC>
-D : <audio resample: 0(off),44100,32000>
-E : <TS buffer(188*n): 1,2,3,4,5,6,7>
-U : <SDI Audio channel: 0(channel 1 2),1(channel 3 4),2(channel 5 6),3(channel 7 8)>
-i : check hdmi input informations:width,height,frames,audio_sample_rate
-c : check Encoder configurations
-r : reboot Encoder
-R : reset Encoder: restore to factory settings
-v : check firmware version
-g : update firmware
-d : enumerate device list
-G : upload logo1.bmp(The width and height of logo must be multiples of 4. For example,64x64,100x64,128x128 and etc.)
-L : <logo enable 0:off,1:on>
-B : <logo alpha: 0(0%),1(25%),2(37.5%),3(50%),4(62.5%),5(75%),6(87.5%),7(100%)>
-x : <x (logo x position)>
-y : <y (logo y position)>
-W : <font enable 0:off,1:on>
-X : <x (font x position)>
-Y : <y (font y position)>
-S : <font size: 8-72>
-H : <font alpha:0(0%),1(25%),2(37.5%),3(50%),4(62.5%),5(75%),6(87.5%),7(100%)>
-C : <font color:0-0xffffffff (red:0xffff0000, green:0xff00ff00,blue:0xff0000ff)>
-M : <Text:up to 255 character>

```

5.1 Here are some examples how to use this command line tool

- **Checking SDI input:**

This includes input resolution, frame rate, and audio information.

```
# ./TBS6324SDITool_x64 -a 0 -i
```

```

root@dw3-All-Series:/home/dw-3/TBS6324SDITool#
root@dw3-All-Series:/home/dw-3/TBS6324SDITool#
root@dw3-All-Series:/home/dw-3/TBS6324SDITool# ./TBS6324SDITool_x64 -a 0 -i
opening device /dev/dvb/adapater0/frontend0 OK!

Stream: ON
Video Info: 1920x1080P60
Audio Sample: 48000

root@dw3-All-Series:/home/dw-3/TBS6324SDITool#
root@dw3-All-Series:/home/dw-3/TBS6324SDITool# █

```

This command is used to check the input information of the first SDI port (-a 0 indicates the first SDI port, where device numbering starts from "0" in a Linux system).

- **Check the current Encoder configuration:**

```
# ./TBS6324SDITool_x64 -a 0 -c
```

```

root@dw3-All-Series:/home/dw-3/TBS6324SDITool#
root@dw3-All-Series:/home/dw-3/TBS6324SDITool# ./TBS6324SDITool_x64 -a 0 -c
opening device /dev/dvb/adapter0/frontend0 OK!

configure information on channel:
Encoded width: auto
Encoded height: auto
Encoding type: H264
H.264 Profile: High
Encoding frame rate: 30
Bitrate control: cbr
cbr Bitrate(K): 6000
Key interval: 60
PMT pid: 260
Transport ID: 264
Video ID: 280
Program ID: 1
Program name: Service 1
TS Null Packet: 0
sdi_audio_channel: channel 1 2
Audio bitrate: 128000
AAC type: LC-AAC
Audio resample: 0
TS buffer: 188x7
Logo enable: 0
Logo X: 100
Logo Y: 200
Logo Alpha: 7
Font enable: 0
Font X: 100
Font Y: 100
Font Alpha: 7
Font Size: 32
Font Color: 0xFFFF0000
Text:

```

Set the output video size (output resolution). For example, if the input is currently 1920x1080_60hz, can either keep the same resolution for the output or encode to a different resolution such as 1280x720. The tool supports scaling. To set the output to 1280x720, use the following command:

```
# ./TBS6324SDITool_x64 -a 0 -s 7
```

```

root@dw3-All-Series:/home/dw-3/TBS6324SDITool# ./TBS6324SDITool_x64 -a 0 -s 7
opening device /dev/dvb/adapter0/frontend0 OK!

root@dw3-All-Series:/home/dw-3/TBS6324SDITool#
root@dw3-All-Series:/home/dw-3/TBS6324SDITool#
root@dw3-All-Series:/home/dw-3/TBS6324SDITool# ./TBS6324SDITool_x64 -a 0 -c
opening device /dev/dvb/adapter0/frontend0 OK!

configure information on channel:
Encoded width: 1280
Encoded height: 720
Encoding type: H264
H.264 Profile: High
Encoding frame rate: 30
Bitrate control: cbr
cbr Bitrate(K): 6000
Key interval: 60
PMT pid: 260
Transport ID: 264
Video ID: 280
Program ID: 1
Program name: Service 1
TS Null Packet: 0
sdi_audio_channel: channel 1 2
Audio bitrate: 128000
AAC type: LC-AAC
Audio resample: 0
TS buffer: 188x7
Logo enable: 0
Logo X: 100
Logo Y: 200
Logo Alpha: 7
Font enable: 0
Font X: 100
Font Y: 100
Font Alpha: 7
Font Size: 32
Font Color: 0xFFFF0000
Text:

```

5.2 Set Frame Rate and Key Interval (Key Frame)

- For example, if the input is currently 1920x1080P60hz, and goal is to encode to 30fps output, can use the following command:
./TBS6324SDITool_x64 -a 0 -f 30 -k 60
- There is a strategy for setting the appropriate "frame rate" and "key interval". For instance, if the SDI input is 1920x1080P60hz, we recommend setting it as follows:
Set the frame rate to 60fps and the key interval to 60/120/180fps.
Alternatively, set the frame rate to 30fps and the key interval to 30/60/90fps.
The principle behind these settings is to keep the frame rate either the same as the input or half of the input, and to set the key interval accordingly.
- Regarding the "Key Interval", we generally recommend setting it to 1x, 2x, or 3x the frame rate. This means that the key frame (I frame) interval can be 1 second, 2 seconds, or 3 seconds."

```

root@dw3-All-Series:/home/dw-3/TBS6324SDITool# ./TBS6324SDITool_x64 -a 0 -f 30 -k 60
opening device /dev/dvb/adapter0/frontend0 OK!

root@dw3-All-Series:/home/dw-3/TBS6324SDITool# ./TBS6324SDITool_x64 -a 0 -c
opening device /dev/dvb/adapter0/frontend0 OK!

configure information on channel:
Encoded width: 1280
Encoded height: 720
Encoding type: H264
H.264 Profile: High
Encoding frame rate: 30
Bitrate control: cbr
cbr Bitrate(K): 6000
Key interval: 60
PMT pid: 260
Transport ID: 264
Video ID: 280
Program ID: 1
Program name: Service 1
TS Null Packet: 0
sdi_audio_channel: channel 1 2
Audio bitrate: 128000
AAC type: LC-AAC
Audio resample: 0
TS buffer: 188x7
Logo enable: 0
Logo X: 100
Logo Y: 200
Logo Alpha: 7
Font enable: 0
Font X: 100
Font Y: 100
Font Alpha: 7
Font Size: 32
Font Color: 0xFFFF0000
Text:

```

5.3 Set Video Codec

- To set the video codec to H.265 (HEVC), use the following command:
./TBS6324SDITool_x64 -a 0 -e H264
./TBS6324SDITool_x64 -a 0 -e H265


```
root@dw3-All-Series:/home/dw-3/TBS6324SDITool# ./TBS6324SDITool_x64 -a 0 -e H265
opening device /dev/dvb/adapater0/frontend0 OK!

root@dw3-All-Series:/home/dw-3/TBS6324SDITool# ./TBS6324SDITool_x64 -a 0 -c
opening device /dev/dvb/adapater0/frontend0 OK!

configure information on channel:
Encoded width: 1280
Encoded height: 720
Encoding type: H265
Encoding frame rate: 30
Bitrate control: cbr
cbr Bitrate(K): 6000
Key interval: 60
PMT pid: 260
Transport ID: 264
Video ID: 280
Program ID: 1
Program name: Service 1
TS Null Packet: 0
sdi_audio_channel: channel 1 2
Audio bitrate: 128000
AAC type: LC-AAC
Audio resample: 0
TS buffer: 188x7
Logo enable: 0
Logo X: 100
Logo Y: 200
Logo Alpha: 7
Font enable: 0
Font X: 100
Font Y: 100
Font Alpha: 7
Font Size: 32
Font Color: 0xFFFF0000
Text:
```

5.4 Set Bitrate Control and Video Bitrate

- For example, to set the bitrate control to "cbr" (constant bitrate) and the video bitrate to 4 Mbps, use the following command:

```
# ./TBS6324SDITool_x64 -a 0 -l cbr -b 4000
```

```
root@dw3-All-Series:/home/dw-3/TBS6324SDITool#
root@dw3-All-Series:/home/dw-3/TBS6324SDITool# ./TBS6324SDITool_x64 -a 0 -l cbr -b 4000
opening device /dev/dvb/adapater0/frontend0 OK!

root@dw3-All-Series:/home/dw-3/TBS6324SDITool# ./TBS6324SDITool_x64 -a 0 -c
opening device /dev/dvb/adapater0/frontend0 OK!

configure information on channel:
Encoded width: 1280
Encoded height: 720
Encoding type: H265
Encoding frame rate: 30
Bitrate control: cbr
cbr Bitrate(K): 4000
Key interval: 60
PMT pid: 260
Transport ID: 264
Video ID: 280
Program ID: 1
Program name: Service 1
TS Null Packet: 0
sdi_audio_channel: channel 1 2
Audio bitrate: 128000
AAC type: LC-AAC
Audio resample: 0
TS buffer: 188x7
Logo enable: 0
Logo X: 100
Logo Y: 200
Logo Alpha: 7
Font enable: 0
Font X: 100
Font Y: 100
Font Alpha: 7
Font Size: 32
Font Color: 0xFFFF0000
Text:
```

5.5 Set PIDs and Program Name

- For example, to set the PMT PID to 178, the Transport PID to 278, the Video PID (Vpid) to 378 (with the Audio PID automatically set to "video pid + 1"), the Service ID (Sid) to 105, and the Program Name to "MYTV", use the following command:

```
# ./TBS6324SDITool_x64 -a 0 -t 178,278,378,105 -N MYTV
```

```
root@dw3-All-Series:/home/dw-3/TBS6324SDITool# ./TBS6324SDITool_x64 -a 0 -t 178,278,378,105 -N MYTV
set <pmt,transpid,vpid,sid> to <178,278,378,105>
set progname to: MYTV
opening device /dev/dvb/adapter0/frontend0 OK!

root@dw3-All-Series:/home/dw-3/TBS6324SDITool# ./TBS6324SDITool_x64 -a 0 -c
opening device /dev/dvb/adapter0/frontend0 OK!

configure information on channel:
Encoded width: 1280
Encoded height: 720
Encoding type: H265
Encoding frame rate: 30
Bitrate control: cbr
cbr Bitrate(K): 4000
Key interval: 60
PMT pid: 178
Transport ID: 278
Video ID: 378
Program ID: 105
Program name: MYTV
TS Null Packet: 0
sdi_audio_channel: channel 1 2
Audio bitrate: 128000
AAC type: LC-AAC
Audio resample: 0
TS buffer: 188x7
Logo enable: 0
Logo X: 100
Logo Y: 200
Logo Alpha: 7
Font enable: 0
Font X: 100
Font Y: 100
Font Alpha: 7
Font Size: 32
Font Color: 0xFFFF0000
Text:
```

5.6 Set Audio Codec and Audio Bitrate

- For example, to set the audio codec to AAC-LC (Low Complexity) and the audio bitrate to 192 kbps, use the following command:

```
# ./TBS6324SDITool_x64 -a 0 -A LC-AA -I 192000
```

5.7 Set Audio Re-sample Rate

- To set the audio re-sample rate to 44.1 kHz, use the following command:

```
# ./TBS6324SDITool_x64 -a 0 -D 44100
```

```
root@dw3-All-Series:/home/dw-3/TBS6324SDITool# ./TBS6324SDITool_x64 -a 0 -D 44100
opening device /dev/dvb/adapter0/frontend0 OK!

sendext: &audio_resample=44100
root@dw3-All-Series:/home/dw-3/TBS6324SDITool# ./TBS6324SDITool_x64 -a 0 -c
opening device /dev/dvb/adapter0/frontend0 OK!

configure information on channel:
Encoded width: 1280
Encoded height: 720
Encoding type: H265
Encoding frame rate: 30
Bitrate control: cbr
cbr Bitrate(K): 4000
Key interval: 60
PMT pid: 178
Transport ID: 278
Video ID: 378
Program ID: 105
Program name: MYTV
TS Null Packet: 0
sdi_audio_channel: channel 1 2
Audio bitrate: 192000
AAC type: LC-AAC
Audio resample: 44100
TS buffer: 188x7
Logo enable: 0
Logo X: 100
Logo Y: 200
Logo Alpha: 7
Font enable: 0
Font X: 100
Font Y: 100
Font Alpha: 7
Font Size: 32
Font Color: 0xFFFF0000
Text:
```

5.8 Set “minQp”, “maxQp”

- MinQP, MaxQP is only working for “vbr” & “evbr” bitrate control.
- MinQP, range 5-48
- MaxQP, should be set higher than “MinQP”, range MinQP~51.
- Normally, the interval between minQP and maxQP should be larger. For example,
set minQP to: 5
maxQP to: 50

5.9 Logo OSD Overlay

- Logo Name and Format:
The logo file should be named "logo1.bmp" and should be in BMP format.
- Logo Size:

The width and height of the logo must be multiples of 4. For example, valid sizes include 64x64 and 128x64.

➤ **Upload Logo File:**

Place the logo file "logo1.bmp" in the same directory as the TBS6324SDITool_x64 tool.

Set Logo Position and Alpha: Use the following command to set the logo position and alpha (transparency) value.

For example, if logo is 64x64 and want to set the position to x: 1800, y: 80, with an alpha of 100% (fully opaque), use the command:

```
# ./TBS6324SDITool_x64 -a 0 -X 1800 -Y 80 -A 255
```

5.10 Turn OSD Logo Overlay On:

➤ **Use the following command to enable the OSD logo overlay:**

```
# ./TBS6324SDITool_x64 -a 0 -L 1
```

```
root@dw3-All-Series:/home/dw-3/TBS6324SDITool# ./TBS6324SDITool_x64 -a 0 -c
opening device /dev/dvb/adapter0/frontend0 OK!

configure information on channel:
Encoded width: auto
Encoded height: auto
Encoding type: H264
H.264 Profile: Main
Encoding frame rate: 30
Bitrate control: cbr
cbr Bitrate(K): 6000
Key interval: 60
PMT pid: 260
Transport ID: 264
Video ID: 280
Program ID: 1
Program name: Service 1
TS Null Packet: 0
sdi_audio_channel: channel 1 2
Audio bitrate: 128000
AAC type: LC-AAC
Audio resample: 0
TS buffer: 188x7
Logo enable: 1
Logo X: 1800
Logo Y: 80
Logo Alpha: 7
Font enable: 0
Font X: 100
Font Y: 100
Font Alpha: 7
Font Size: 32
Font Color: 0xFFFF0000
Text:
```

5.11 Text OSD Overlay

➤ To set up and enable a text OSD overlay, follow these steps:

Set Text Location and Font Size: Use the following command to specify the location (X and Y coordinates), font size, and text content. For example, to set the text location to X: 1000, Y: 1000, font size to 20, and text content to "welcome to my channel," use the command:

```
# ./TBS6324SDITool_x64 -a 0 -X 1000 -Y 1000 -S 20 -H 7 -M "welcome to my channel"
```

5.12 Enable Text OSD Overlay

- Use the following command to enable the text OSD overlay:
./TBS6324SDITool_x64 -a 0 -W 1

```
root@dw3-All-Series:/home/dw-3/TBS6324SDITool# ./TBS6324SDITool_x64 -a 0 -c
opening device /dev/dvb/adapter0/frontend0 OK!

configure information on channel:
Encoded width: auto
Encoded height: auto
Encoding type: H264
H.264 Profile: Main
Encoding frame rate: 30
Bitrate control: cbr
cbr Bitrate(K): 6000
Key interval: 60
PMT pid: 260
Transport ID: 264
Video ID: 280
Program ID: 1
Program name: Service 1
TS Null Packet: 0
sdi_audio_channel: channel 1 2
Audio bitrate: 128000
AAC type: LC-AAC
Audio resample: 0
TS buffer: 188x7
Logo enable: 0
Logo X: 1800
Logo Y: 80
Logo Alpha: 7
Font enable: 1
Font X: 1000
Font Y: 1000
Font Alpha: 7
Font Size: 20
Font Color: 0xFFFF0000
Text: wellcome to my channel
```

5.13 Reboot Encoder

- To reboot the encoder, use the following command:
./TBS6324SDITool_x64 -a 0 -r
The reboot process typically takes about 30 seconds.

5.14 Reset Encoder

- To reset the encoder to its factory settings and reboot it, use the following command:

```
# ./TBS6324SDITool_x64 -a 0 -R
```

The reset process, which includes both factory reset and reboot, typically takes about 40 seconds.

5.15 Check Encoder Firmware Version

- To check the firmware version of the encoder, use the following command:

```
# ./TBS6324SDITool_x64 -a 0 -v
```

```
root@dw3-All-Series:/home/dw-3/TBS6324SDITool#
root@dw3-All-Series:/home/dw-3/TBS6324SDITool# ./TBS6324SDITool_x64 -a 0 -v
opening device /dev/dvb/adapter0/frontend0 OK!

version: v4_20240929_NSDI_OEMF_release

root@dw3-All-Series:/home/dw-3/TBS6324SDITool#
root@dw3-All-Series:/home/dw-3/TBS6324SDITool#
```

5.16 Upgrade Encoder Firmware

- To upgrade the firmware of the encoder, follow these steps:
Place the firmware file named upgrade.bin in the same directory as the TBS6324SDITool_x64 setting tool.
Use the following command to upload and upgrade the firmware:

```
# ./TBS6324SDITool_x64 -a 0 -g
```

```
root@dw3-All-Series:/home/dw-3/TBS6324SDITool# ls
bin  dvblast.sh  logol.bmp  log-sdi  readme  src  TBS6324SDITool_x64  upgrade.bin  za
root@dw3-All-Series:/home/dw-3/TBS6324SDITool#
root@dw3-All-Series:/home/dw-3/TBS6324SDITool#
root@dw3-All-Series:/home/dw-3/TBS6324SDITool#
root@dw3-All-Series:/home/dw-3/TBS6324SDITool#
root@dw3-All-Series:/home/dw-3/TBS6324SDITool# ./TBS6324SDITool_x64 -a 0 -g
```

Afterwards, need to reboot the encoder to load the newly upgraded firmware.

6. Streaming Media Transmission on Linux

- This PCIe Encoder card is built on the DVB (Digital Video Broadcasting) Architecture and is compatible with DVB streaming software. Examples of such software include DVblast, Tvheadend, Astra Cesbo, mumuDVB, and others. Simply treat it as a DVBS/S2 tuner card. This means that when tuning a channel, need to configure parameters such as 'frequency', 'polarization', 'symbol rate', and 'modulation'.
- When configuring these parameters, can set any 'fake' values, but they must comply with

the DVBS/S2 standard. For example:

Frequency: Configure values such as 1253800 or 4732000.

Polarization: Choose either V (Vertical) or H (Horizontal).

Symbol rate: Possible values include 20000000, 41250000, but note that 9870 is not a valid symbol rate and should be corrected.

Modulation: Configure 'QPSK' or '8PSK'.

- Please ensure that the values configure are within the valid ranges and comply with the DVBS/S2 standard to ensure proper operation.

6.1 DVB last Streaming

- Support RTP/UDP Unicast & Multicast

6.1.1 Tuning a channel:

- To tune to a specific channel using dvblast, execute the following command with the appropriate parameters:

```
# dvblast -f 12538000 -s 45000000 -v 13 -a 0
```

```
root@dw3-All-Series:/home/dw-3/TBS6324SDItool# dvblast -f 12538000 -s 45000000 -v 13 -a 0
DVblast 3.4 (release)
warning: restarting
debug: compiled with DVB API version 5.11
debug: using DVB API version 5.12
debug: Frontend "TurboSight TBS 6324 3G-SDI PCIe Encoder" supports:
debug: frequency min: 950000, max: 2150000, stepsize: 0, tolerance: 0
debug: symbolrate min: 1000000, max: 45000000, tolerance: 0
debug: capabilities:
debug: INVERSION_AUTO
debug: FEC_1_2
debug: FEC_2_3
debug: FEC_3_4
debug: FEC_4_5
debug: FEC_5_6
debug: FEC_6_7
debug: FEC_7_8
debug: FEC_AUTO
debug: QPSK
debug: 2G_MODULATION
debug: FE_CAN_RECOVER
debug: delivery systems:
debug: DVBS
debug: DVBS2
debug: frequency 12538000 is in Ku-band (higher)
debug: configuring LNB to v=13 p=0 satnum=0 uncommitted=0 lnb-type=universal bis frequency=1938000
debug: tuning DVB-S frontend to f=12538000 srate=45000000 inversion=-1 fec=999 rolloff=35 modulation=legacy pilot=-1 mis=0 /pls-mode: ROOT (0) pls-code: 0 is-id: 0 /
warning: failed opening CAM device /dev/dvb/adapter0/ca0 (No such file or directory)
debug: setting filter on PID 0
debug: setting filter on PID 16
debug: setting filter on PID 17
debug: setting filter on PID 18
debug: setting filter on PID 19
debug: setting filter on PID 20
error: no config file
debug: frontend has acquired signal
debug: frontend has acquired carrier
debug: frontend has acquired stable FEC
debug: frontend has acquired sync
info: frontend has acquired lock
lock status: 1
debug: - Bit error rate: 0
debug: - Signal strength: 62940
debug: - SNR: 26896
debug: new SDT actual tsid=278 version=6 onid=65281
debug: * service sid=105 running=4
debug: - desc 48 service type=0x1 provider="Encoder Pro" service="Service 1"
debug: end SDT
debug: new PAT tsid=278 version=6
debug: * program number=105 pid=178
debug: end PAT
```

6.1.2 Stream

- To set up streaming with dvblast, need to create a configuration file with a "*.conf" extension. This file specifies the channel to be streamed, as well as the output address and port.
- For instance, create a configuration file named "dvb.conf" with the following content:
239.255.0.1:1234/udp 1 105

Here's a breakdown of the contents:

- 239.255.0.1:
This is the output address, a multicast IP address used for sending data to multiple recipients simultaneously within a local network.
1234: This is the output port, the number through which the data will be transmitted over the network. The receiving end needs to listen on this port to receive the data.
- Explanation of DVBLast Configuration:
"always on" Flag: This flag indicates to DVBLast whether the configured channel should remain active at all times (i.e., continuously transmitting) or if it may experience interruptions.
- Program Number 105:
In this configuration, the number "105" represents a specific program number. In Digital Video Broadcasting (DVB) environments, each program is assigned a unique number for identification and selection during transmission.
- To initiate streaming with dvblast, use a command similar to this:
dvblast -f 12538000 -s 45000000 -v 13 -a 0 -c dvb.conf
- The output URL will be: udp://@239.255.0.1:1234
This is streaming out in UDP multicast.
- UDP Unciast, "dvb.conf":
192.168.1.168:1235/udp 1 105
URL: udp://@192.168.1.168:1235
- RTP Multicast, "dvb.conf":
239.255.0.2:1236 1 105
URL: rtp://239.255.0.2:1236
- RTP Unicast, "dvb.conf":
192.168.1.169:1237 1 105

URL: [rtp://192.168.1.169:1237](http://192.168.1.169:1237)

- For more detailed information on how to use dvblast, please refer to the following link:
<https://github.com/videolan/dvblast/blob/master/README>

6.2 Tvheadend Streaming Setup

6.2.1 Install Tvheadend

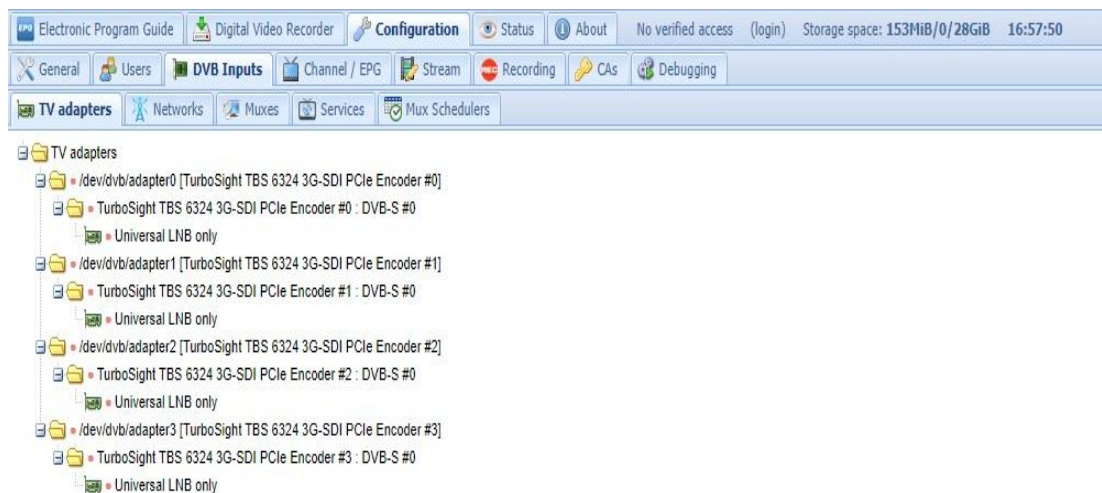
- Install Tvheadend by either compiling the source code from its official repository or using a package manager like snap if it is available in system's "snap store".

6.2.2 Start Tvheadend

- To start the Tvheadend service, typically use a command like this:
`# tvheadend -C`

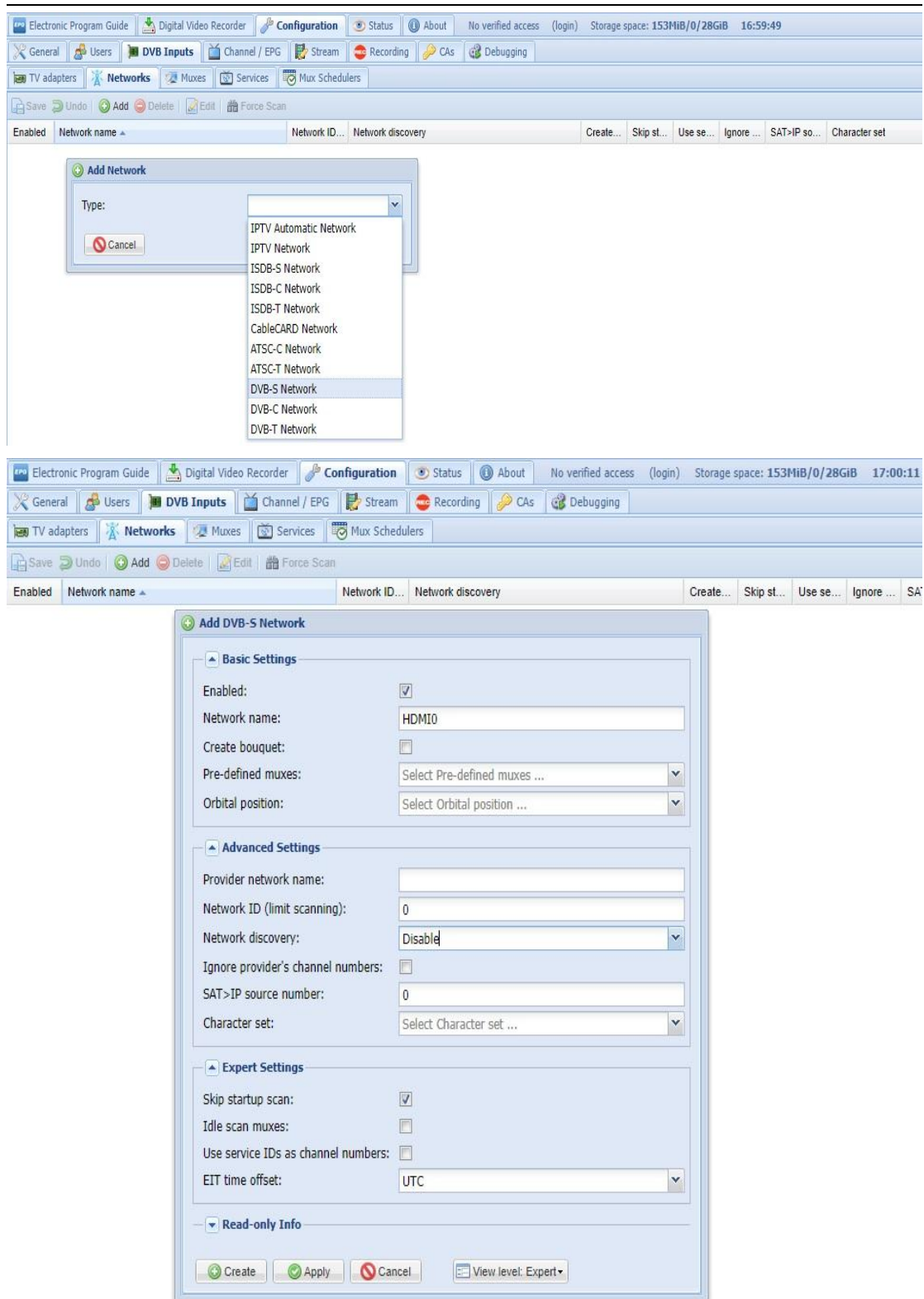
6.2.3 Login to Tvheadend Web Interface

- After starting Tvheadend, access its web interface to configure and manage streaming services. The default URL for the web interface is usually:
<http://ip:9981>

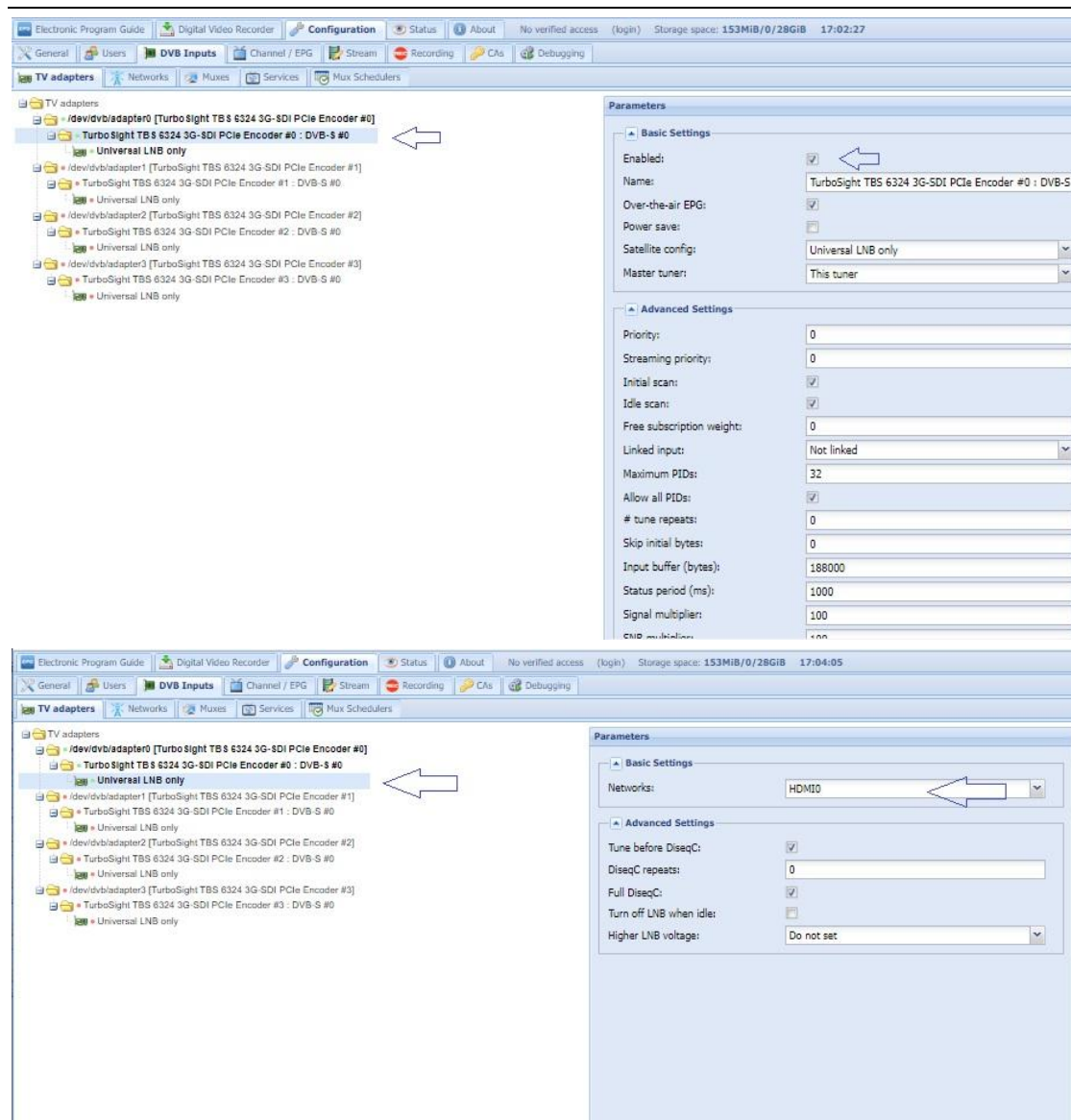


6.2.4 Tune a Channel

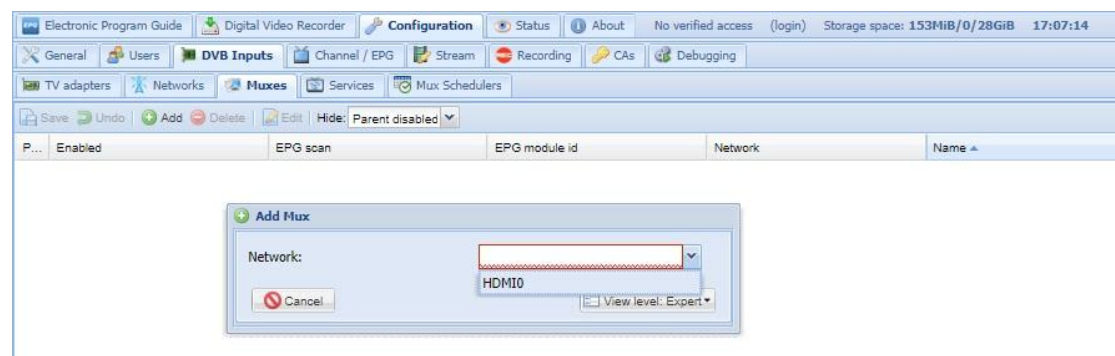
- Create a "DVBS" Network

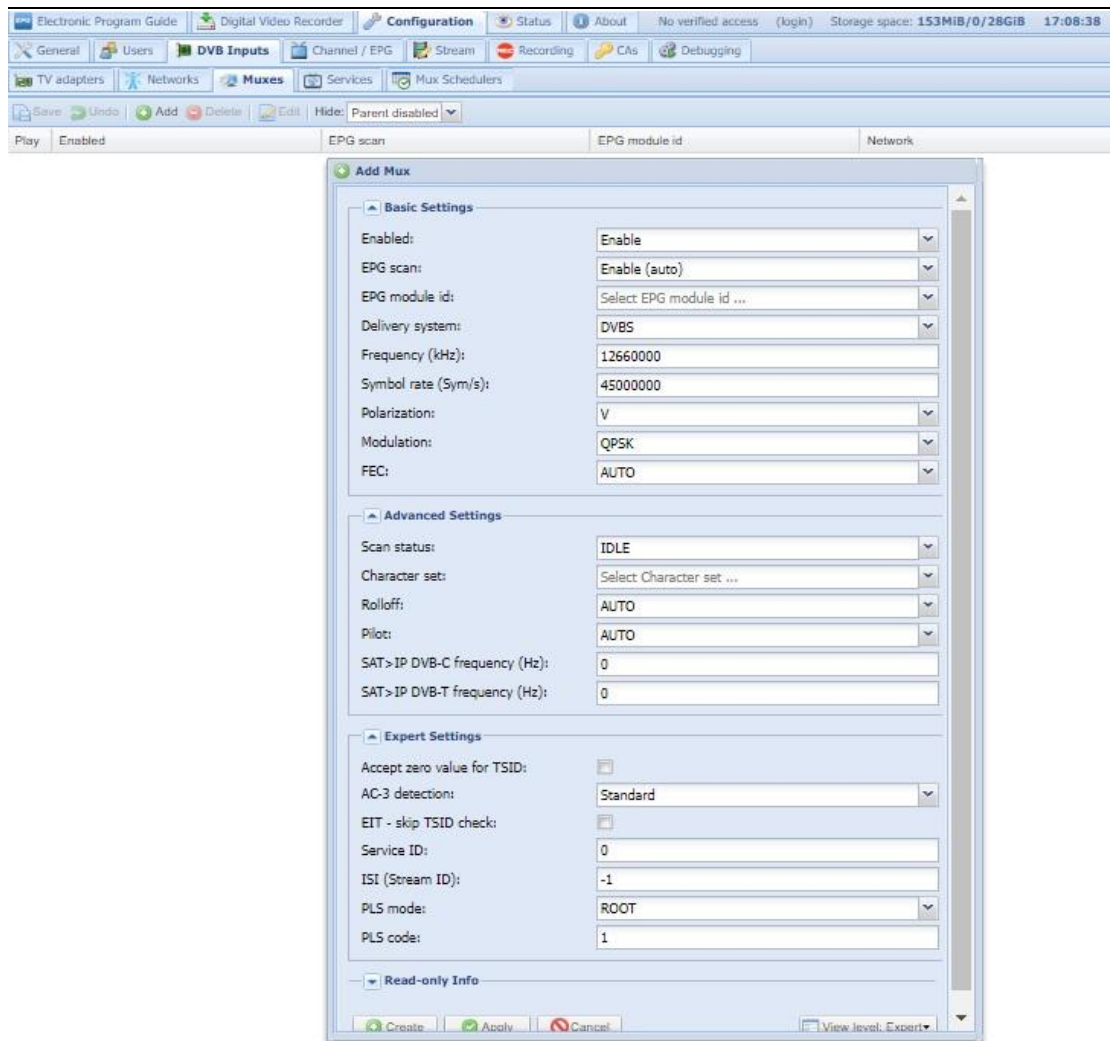


- Enable the Adapter device, and assign the “Network” to the Adapter



- **Muxes:**
To configure the PCIe Encoder for DVBS/S2 standard, set the “frequency”, “polarity (pol.)”, “symbol rate”, and “modulation”.





- Click on “Create” and wait a few seconds. If the signal is locked, will see “lock status is ok” and a list of “services” like this:

EPG scan	EPG module id	Network	Name	Orig...	Tran...	Scan status	Scan result	Character set	Services	M
Enable (auto)		HDMI0	12660V	85281	278	IDLE	OK		1	0

- Services:
Map the channel (or service) out.

P...	D...	E...	Channel	Service name	Map selected services	Prio...	E...	Type override	Network	Mux
			Mytv		Map all services	0		Override disabled	HDMI0	12660V

- Channel/EPG:
Assign a number to the channel. For example:



- If assign the "Number" as 10, then the streaming URL will be formatted as:
`http://ip:9981/stream/channelnumber/10`
Using Linux server's IP address as an example, if server's IP is 192.168.8.47 and assigned the channel number 10, then the URL will be:
`http://192.168.8.47:9981/stream/channelnumber/10`

6.3 Astra Cesbo Streaming

6.3.1 Install Astra Cesbo

- Astra Cesbo provides a "Binary" file instead of source code for installation.
`wget https://cesbo.com/astra-latest`
`chmod +x astra-latest`
`cp astra-latest /usr/bin/astra`

6.3.2 Create a Folder for Astra Cesbo

- Create a dedicated folder to store Astra Cesbo's configurations and related files.
`mkdir -p /etc/astra`

6.3.3 Create a Configuration File for Astra Cesbo

- Create a JSON file to store Astra Cesbo's configurations:
`vi /etc/astra/astra.conf`

6.3.4 License

- To use Astra Cesbo, need to purchase a license or apply for a trial license from Astra Cesbo. Once have obtained the license, place the license file named `license.txt` in the `/etc/astra/` directory.
Purchase a license or apply for a trial license by visiting:
`https://app.cesbo.com/orders/software/astra/buy/`

6.3.5 Start Astra Cesbo

- To start Astra Cesbo, use the following command:
`# astra -c /etc/astra/astra.conf -p 6916 -daemon`
- This command will start Astra Cesbo with the configuration file located at `/etc/astra/astra.conf`, listening on port 6916, and running in the background as a daemon.
- Also, verify whether Astra Cesbo has started by using the following command:
`# ps -aux | grep astra`

6.3.6 Logging into Astra Cesbo Web GUI

- To access the Astra Cesbo web graphical user interface (GUI), use the following URL:
`http://ip:6916`
- Replace `<ip>` with the IP address of Linux server. For example, if server's IP address is 192.168.8.47, then the login site will be:
`http://192.168.8.47:6916`
The default login credentials are: admin/admin.

192.168.8.47:6916/#/

admin

☐ Remember me

[SIGN IN](#)

[Dashboard](#) [Sessions](#) [Settings](#) [Log](#) [Help](#) Click or press 'S' to search...

[New Adapter](#) [New Stream](#) [View](#)

For security reasons, please, change default password!
[CHANGE PASSWORD](#)

6.3.7 New Adapter: Tuning a Channel

General

☒ Enable

NAME
channel-0

ID

VIRTUAL

☐ SAT>IP

ADAPTER
0.0 : TurboSight TBS 6324 3G-SDI PCIe Encoder

Available

0.0 : TurboSight TBS 6324 3G-SDI PCIe Encoder
1.0 : TurboSight TBS 6324 3G-SDI PCIe Encoder
2.0 : TurboSight TBS 6324 3G-SDI PCIe Encoder
3.0 : TurboSight TBS 6324 3G-SDI PCIe Encoder

APPLY

SCAN

CANCEL

General

☒ Enable

NAME
channel-0

ID

VIRTUAL

☐ SAT>IP

ADAPTER
0.0 : TurboSight TBS 6324 3G-SDI PCIe Encoder

TYPE

Satellite

DVB-S
DVB-S2

Terrestrial

DVB-T
DVB-T2
ATSC
ISDB-T

Cable

DVB-C
DVB-C (Annex A)
DVB-C (Annex B)
DVB-C (Annex C)

- To tune the PCIe Encoder card to a channel in the DVBS/S2 standard, configure the "frequency," "polarization" (pol.), and "symbol rate."

[General](#) [LNB](#) [Advanced](#)

☒ Enable

NAME

ID

VIRTUAL

☐ SAT-IP

ADAPTER

TYPE

TRANSPONDER

FREQUENCY

POLARIZATION

SYMBOLRATE

[APPLY](#) [SCAN](#) [CANCEL](#)

- To tune the PCIe Encoder card to a channel in the DVBS/S2 standard, configure the "frequency," "polarization" (pol.), and "symbol rate."

Dashboard Sessions Settings Log Help Click or press 'S' to search...

channel-0

95%

95%

40%

[General](#) [LNB](#) [Advanced](#)

☒ Enable monitoring

MODULATION

PCS

STREAM ID

TIMEOUT

DIOCI

CI

CI DEVICE

CI RATE

FE DEVICE

BISS

☒ DVB API v3

☐ Budget Mode: Disable hardware PID filtering

RECEIVING_BUFFER_SIZE

CA_DELAY

[APPLY](#) [SCAN](#) [CANCEL](#)

- Click on "SCAN":

Dashboard Sessions Settings Log Help Click or press 'S' to search...

channel-0

95%

95%

40%

[General](#) [LNB](#) [Advanced](#)

SIGNAL

CARRIER

REC

SYNC

LOCK

BER:0

UNC:0

S:0.00dBm

CIN:0.00dB

12.50Mbps

56%

40%

☒ Full: 105 Mhz
 VFD: 375 MPEG-4
 APID: 375

☐ Set DVB-CI CAM for selected channels

[APPLY](#) [CANCEL](#)

- Once the channel has been scanned, select the desired channel and click "Apply."



- Editing a Channel and Configuring Streaming

General Groups Service Remap Backup EPG Advanced

☒ Enable

NAME: Mytv

ID: a002

☐ Create Multi Program Stream

DESCRIPTION:

INPUT LIST NEW INPUT

#1: dvb://a001#pnr=105 ↑↓ ⚙

OUTPUT LIST NEW OUTPUT

#1: ⚙

☐ Remove

APPLY CLONE ANALYZE CANCEL

☒ Enable

OUTPUT TYPE: ▼

- SRT
- HTTP/HLS
- UDP
- RTP
- RES1 Modulator
- NetworkPush
- MPEG-TS Files

- When editing a channel, configure the following streaming options:

HTTP out:

192.168.8.47:6916/#/

Log Help Click or press 'S' to search...

☒ Enable

OUTPUT TYPE: HTTP/HLS ▼

ADDRESS: 192.168.8.47

PORT: 6909

PATH: /mychannel/tv01

☐ Remove

OK CANCEL

General
Groups
Service
Remap
Backup
EPG
Advanced

☒ Enable

NAME
Mytv

ID
a002

☐ Create Multi Program Stream

DESCRIPTION

INPUT LIST
NEW INPUT

#1
dvb://a001#pnr=105

OUTPUT LIST
NEW OUTPUT

#1
http://192.168.8.47:6909/mychannel/tv01

☐ Remove

APPLY
CLONE
ANALYZE
CANCEL

Dashboard Sessions Settings Log Help Click or press 'S' to search...



➤ UDP out:

☒ Enable

OUTPUT TYPE
UDP

LOCAL INTERFACE

enp2s0: 192.168.8.45
Default: Use system route
enp2s0: 192.168.8.45
lo: 127.0.0.1
enp1s0

Destination Port: Default: 1234

☐ Remove

OK
CANCEL

☒ Enable

OUTPUT TYPE
UDP

LOCAL INTERFACE
enp2s0: 192.168.8.45

ADDRESS
239.255.4.3

PORT
4322

☐ Remove

OK
CANCEL

General
Groups
Service
Remap
Backup
EPG
Advanced

☒ Enable

NAME
Mytv

ID
a002

☐ Create Multi Program Stream

DESCRIPTION

INPUT LIST
NEW INPUT

#1
dvb://a001#pnr=105

OUTPUT LIST
NEW OUTPUT

#1
udp://enp2s0@239.255.4.3:4322

☐ Remove

APPLY
CLONE
ANALYZE
CANCEL

➤ RTP output:

☒ Enable

OUTPUT TYPE
RTP

LOCAL INTERFACE
enp2s0: 192.168.8.45

ADDRESS
239.255.7.8

PORT
3728

☐ Remove

OK
CANCEL

➤ SRT output:

Listener mode

An easy way to set up the listener mode is to use VLC player as the "Caller" and the Linux server as the "Listener." For instance, if want laptop's VLC player to call Linux server (assuming laptop's IP is 192.168.8.43), the URL use would be: "srt://192.168.8.47:6205."

☒ Enable

OUTPUT TYPE
 SRT

SRT MODE
 Listener

LOCAL INTERFACE
 Default: Any interface

PORT
 6205

TIMEOUT
 Default: 5

LATENCY
 Default: 120

PASSPHRASE

CRYPTO KEY LENGTH
 Default: 16 AES-128

☐ Timestamp-based packet delivery mode

☐ Remove

OK CANCEL

➤ Caller mode

In caller mode, the Linux server can initiate a call to another device, such as laptop's Vmix software (assuming laptop's IP is 192.168.8.43).

☒ Enable

OUTPUT TYPE
 SRT

SRT MODE
 Caller

ADDRESS
 192.168.8.43

PORT
 2812

TIMEOUT
 Default: 5

LATENCY
 100

STREAM ID

PASSPHRASE

CRYPTO KEY LENGTH
 Default: 16 AES-128

☐ Timestamp-based packet delivery mode

☐ Remove

OK CANCEL

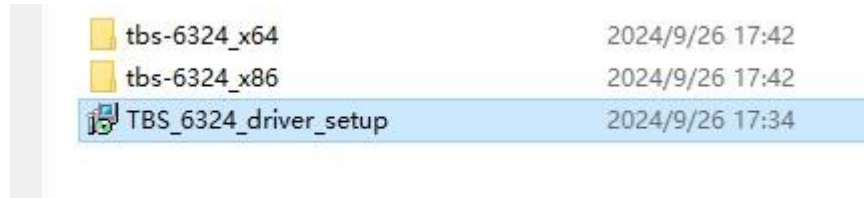
For more detailed instructions on how to configure Astra Cesbo, please refer to the following links:

<https://help.cesbo.com/>

<https://www.cesbo.com/>

7. Usage on Windows Systems

7.1 Install the Windows driver



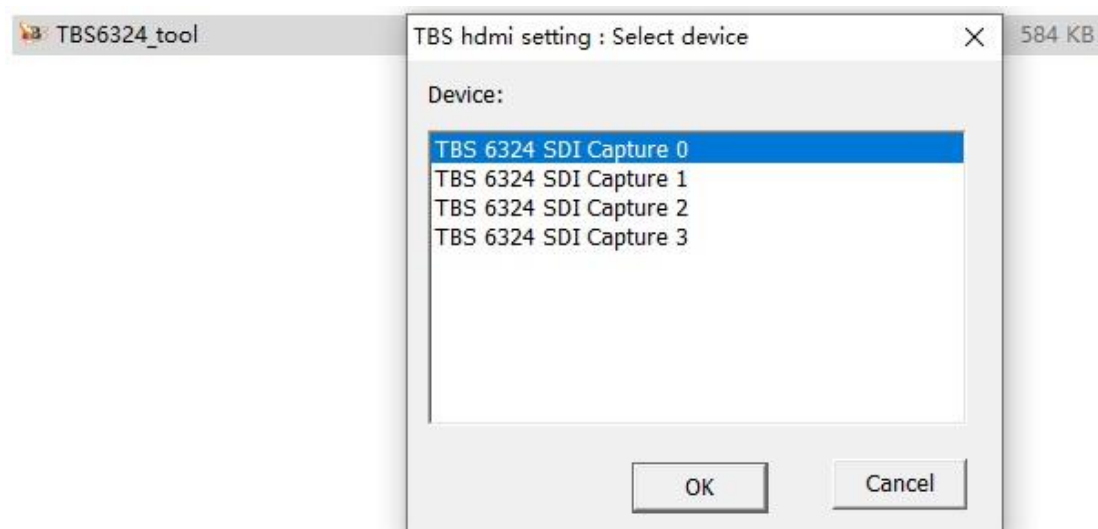
7.2 TBS6324 Device Management

- On computer, navigate to Device Manager and locate the Audio, Video & Game Controllers section to find TBS6324 device.



7.3 Encode Settings

- We provide a GUI (Graphical User Interface) tool for configuring the TBS6324 SDI PCIe Encoder in the Windows system.



TBS6324_tool -- TBS 6324 SDI Capture 0

Input Format:

Output Format:

Encoding Type: H264	Encoded frame rate: 30 [5-60]
Bitrate Control: cbr	key interval: 60 [1-180]
Encoded size: auto	BitRate(K): 6000 [160-20000]
H.264 Profile: Main	MinQp: 5 [5-48]
TS Null Packet: OFF	MaxQp: 32 [MinQp-51]
Audio Bitrate: 128000	PMT ID: 260 [16-7936]
Audio Resample: OFF	Transport ID: 264 [16-3840]
AAC Codec: LC-AAC	Video pid: 280 [16-3840]
TS buffer: 188x7	Program ID: 1
SDI Audio channel: channel 1 2	Program name: HDMI0

Input Format: This indicates the SDI input resolution and frame rate.

Get Input: Use this to check the SDI input information, including resolution and frame rate.

V4_20240929: This is the encoder firmware version.

Set Format: Once have completed the encode configurations, click this button to activate settings.

Reboot: This option allows to reboot the encoder.

Reset: This will perform a factory reset and then reboot the encoder.

Upgrade: Use this to upgrade the firmware of the encoder.

7.4 Tuning Channels in Windows

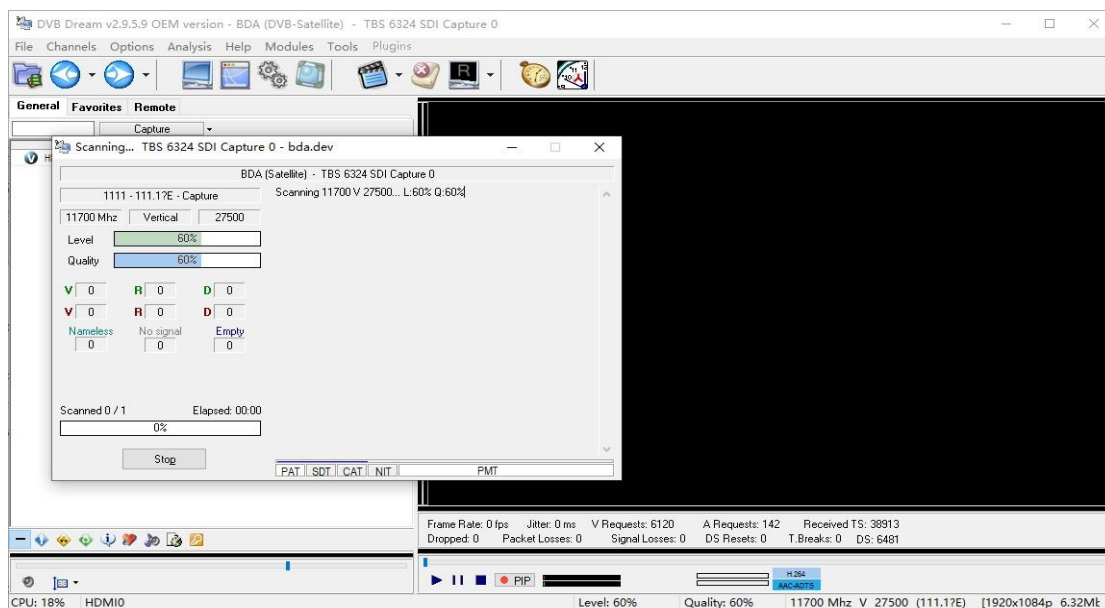
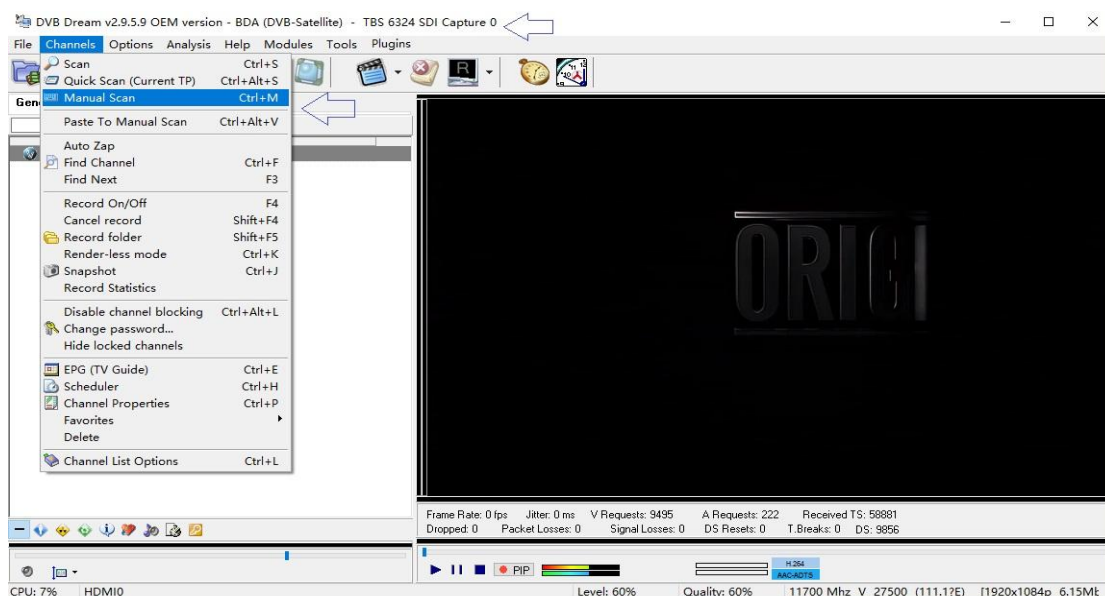
- This SDI PCIe Encoder is built on the DVB architecture and is compatible with Windows DVB

software, such as DVBdream, DVbviewer, Tsreader, TransEdit, and others.

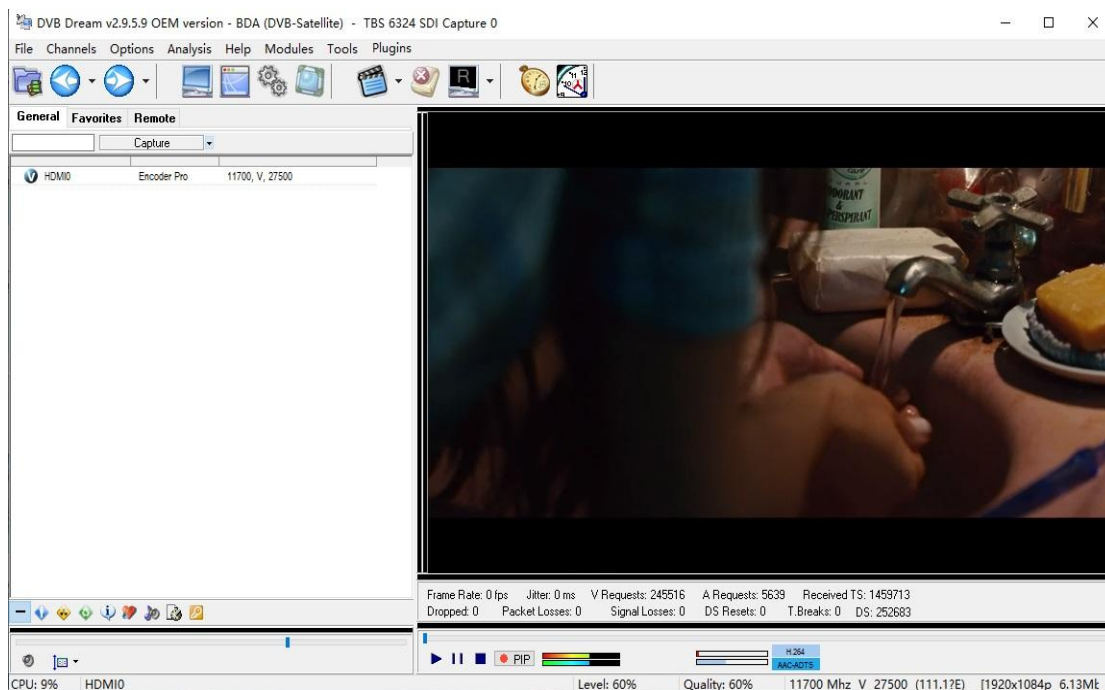
- When tuning channels, simply treat the TBS6324 as a DVBS/S2 tuner card and configure the "Frequency," "Polarization," "Symbol Rate," and "Modulation."
- For "Frequency" and "Symbol Rate," can set any values that comply with the DVBS/S2 standard. "Polarization" can be either V (vertical) or H (horizontal). For example, tune to "12721000, V, 41250000, QPSK."

Here's an example of how to tune channels in DVBdream:

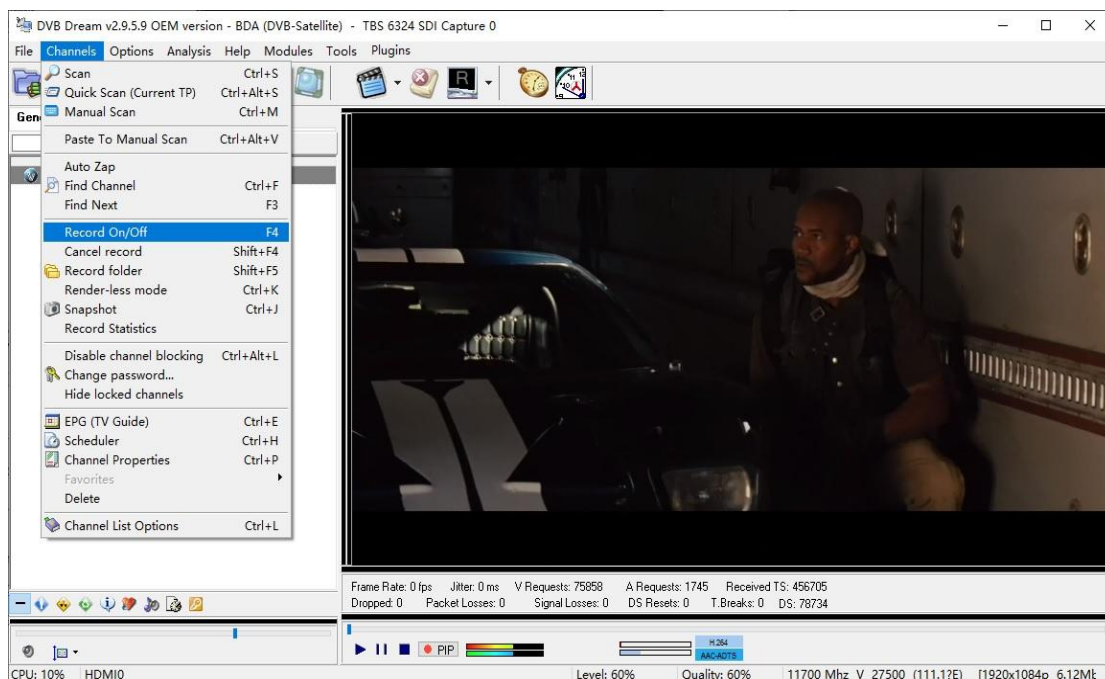
- Device Selection: Choose "Tuner 0" in DVBdream's Options menu under the Device section.
- Manual Scan: Navigate to the Channels section and select Manual Scan.



- Lock Status: Once a channel is locked, proceed with further actions.

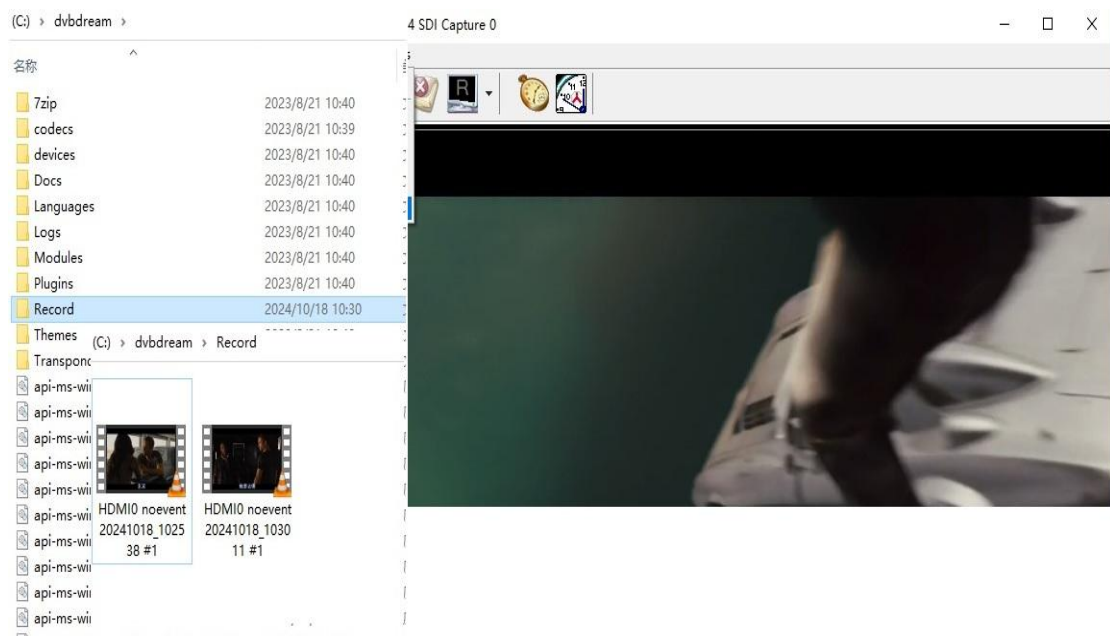


- Recording in DVBDream (supports *.ts format):
First Method:
Toggle recording on or off by selecting "Record on/off" in the Channels section.

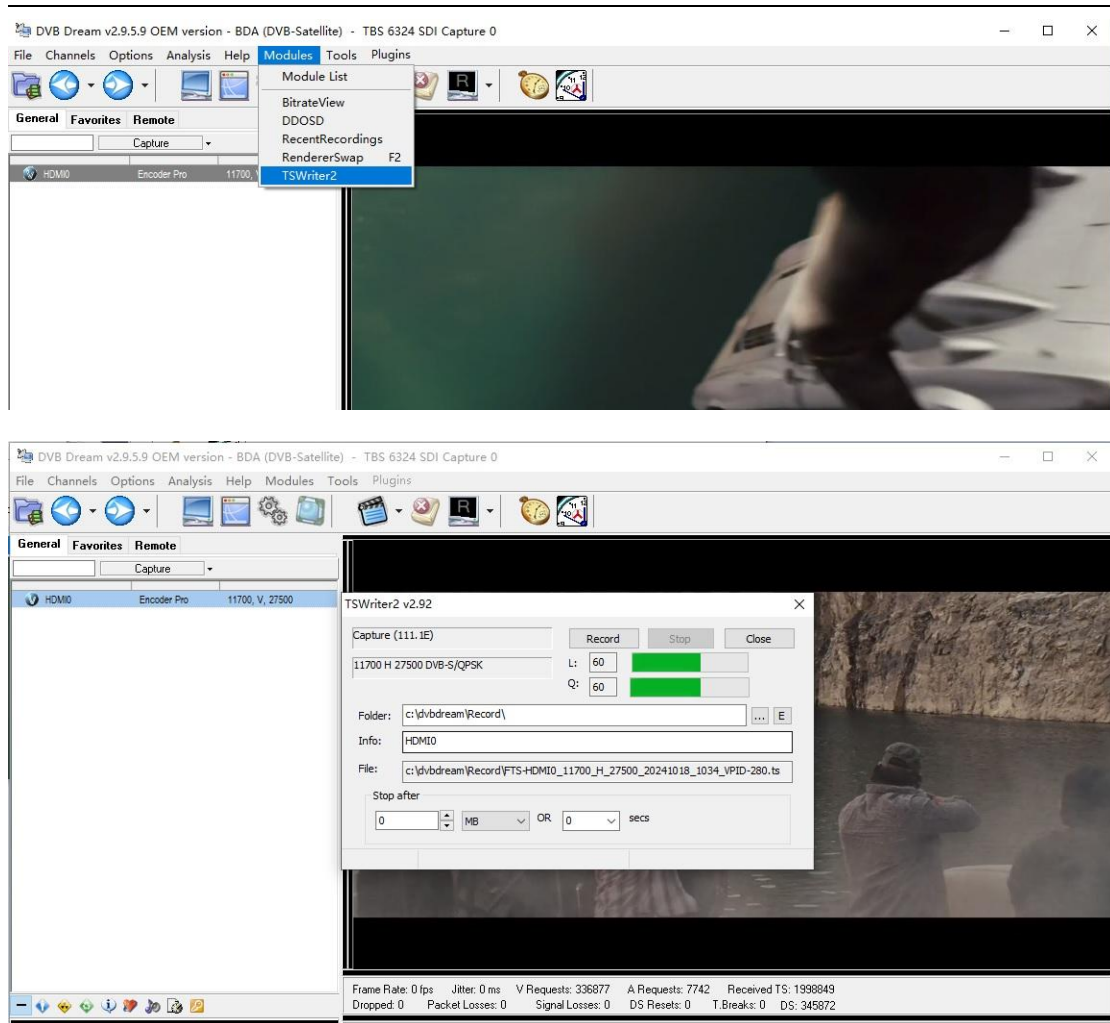




- Toggle recording on or off by selecting "Record on/off" in the Channels section.
- The recorded file will be saved in the DVBDream software installation directory.



- **Second Method:**
Use the TSWriter2 module in DVBDream for recording.



- Other software that supports recording includes DVBCViewer, TransEdit, and TSReader. For TSReader to work with the TBS6324 PCIe Encoder card, may need the TBS6324 *.dll file. Please contact us to obtain these *.dll files.

7.5 Streaming in Windows

- TSReader and TransEdit support the UDP protocol for streaming. However, if streaming is primary use case, Linux may be a better choice due to its wide variety of streaming software. Windows DVB software is primarily designed for watching and recording.

8. Maintenance

8.1 Instructions for returning by mail

- Instruments returned for repair or calibration, either within or out of the warranty period, should be sent with the following information: Name of the Company, name of the contact person, address, telephone number, receipt (in the case of coverage under warranty) and a description of the problem or the service required.
- Mail: support@tbsdtv.com or support@tbsiptv.com.

8.2 Cleaning Recommendations

- **CAUTION:** To clean the cover, take care the instrument is disconnected.
- **CAUTION:** Do not use scented hydrocarbons or colorized solvents. The cover should be cleaned by means of a light solution of detergent and water applied with a soft cloth. Dry thoroughly before using the system again.
- **CAUTION:** Do not use for the cleaning alcohol or its derivatives, these products can attack the mechanical properties of the materials and diminish their useful time of life.